

Jack Liell-Cock

D.Phil. Candidate in Computer Science
University of Oxford

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Education

- 2022 – now  **D.Phil. in Computer Science**, *University of Oxford*, UK
Supervisors: Sam Staton and Jeremy Gibbons
Scholarship: Clarendon Scholar
Expected completion: October 2026
Teaching and Tutoring: Principles of Programming Languages, Functional Programming, Imperative Programming 1 and 2, Design and Analysis of Algorithms, Models of Computation
- 2015 – 2019  **B.Eng. Hons. in Mechatronics (Space)**, *University of Sydney*, Australia
Grade: First Class Honours and University Medal
Thesis: Preserving Privacy of the Influence Structure in Distributed Systems
Supervisor: Ian Manchester
- 2015 – 2019  **B.Sc. in Advanced Mathematics**, *University of Sydney*, Australia
Grade: High Distinction

Publications

Authors of the theoretical computer science papers are listed alphabetically, following the field's convention.

-  O. Kammar, **J. Liell-Cock**, S. Lindley, C. Matache, and S. Staton, “An equational axiomatization of dynamic threads via algebraic effects,” in *ACM Symposium on Principles of Programming Languages (POPL)*, **Distinguished Paper**, 2026. DOI: 10.1145/3776706.
-  **J. Liell-Cock**, Z. Shirazi, and S. Staton, “The relative monadic metalanguage,” in *ACM Symposium on Principles of Programming Languages (POPL)*, 2026. DOI: 10.1145/3776702.
-  **J. Liell-Cock** and S. Staton, “Imprecise probabilistic programming, precisely (functional pearl),” in *ACM International Conference on Functional Programming (ICFP)*, 2026. DOI: 10.1145/3828698.
-  **J. Liell-Cock** and S. Staton, “Compositional imprecise probability,” in *ACM Symposium on Principles of Programming Languages (POPL)*, 2025. DOI: 10.1145/3704890.
-  **J. Liell-Cock** and T. Schrijvers, “Let a thousand flowers bloom: An algebraic representation for edge graphs,” *The Art, Science, and Engineering of Programming*, 2024. DOI: 10.22152/programming-journal.org/2024/8/9.
-  **J. Liell-Cock**, I. Manchester, and G. Shi, “Preserving privacy of the influence structure in Friedkin Johnsen systems,” in *IEEE Conference on Decision and Control (CDC)*, 2020. DOI: 10.1109/CDC42340.2020.9304100.
-  Y. Mao, **J. Liell-Cock**, S. Douglas, *et al.*, “TweetS@: A 3D printed tweeting cubesat with a neural network sun sensor,” in *Australian Space Research Conference*, 2019.

Employment

- 2024 – 2027  **Research Assistant**, *ARIA Safeguarded AI Programme*, Oxford, UK
- *Grant Title*: Employing Categorical Probability towards Safe AI
 - *Principal Investigator*: Sam Staton
 - Building theoretical frameworks for world models with epistemic uncertainty
- 2020 – 2022  **Quant Trader Developer**, *Tibra Capital*, Austinmer, Australia
- Developed an automated strategy trading index derivatives
 - Built a backtesting framework for strategy evolution
 - Trained graduates in research, programming and trading skills
- 2016 – 2019  **Academic Area Tutor**, *St Andrews College*, Sydney, Australia
- Tutored students in mathematics, physics and engineering
 - Mentored students struggling to meet academic requirements
 - Presented relevant and stimulating research fields to the college body

Software

-  **Lead Developer**, *The Cambria Language*
- A language for parametrized algebraic effects and handlers
 - Playground at <https://cambria-lang.org/>
 - Implementation at <https://github.com/jacklc3/cambria>
-  **Author & Maintainer**, *imp-ppl*
- A Haskell DSL for imprecise probabilistic programming via BDDs
 - Package at <https://hackage.haskell.org/package/imp-ppl>
 - Implementation at <https://github.com/jacklc3/imp>
-  **Author & Maintainer**, *algebraic-edge-graphs*
- A Haskell library for algebraic edge-graph construction and transformation
 - Package at <https://hackage.haskell.org/package/algebraic-edge-graphs>
 - Implementation at <https://github.com/jacklc3/algebraic-edge-graphs>

Academic Service

- Dec. 2026  **Co-organizer**, *Autonomous Machine Learning Research (AutoMLR)*
Workshop at NeurIPS 2026, Sydney, Australia
- Dec. 2025  **Program Committee**, *Epistemic Intelligence in Machine Learning (EIML)*
Workshop at EurIPS 2025, Copenhagen, Denmark
- Ongoing  **Reviewer**
Venues including LICS, LMCS and JFP

Internships

- Mar. 2024 – Jun. 2024  **NII International Internship Program**, *National Institute of Informatics*, Tokyo, Japan
Topic: Handlers of Parametrized Algebraic Effects
Supervisor: Taro Sekiyama
- Nov. 2018 – Feb. 2019  **Tibra Summer Internship**, *Tibra Capital*, Austinmer, Australia
Researched and developed high-frequency trading strategies (Proprietary)
- Dec. 2017 – Feb. 2018  **Summer Research Scholarship**, *University of Sydney*, Sydney, Australia
Topic: Automated Refereeing in Football
Supervisor: Abbas El-Zein
- Developed a smart soccer ball to aid referees in detecting offsides
 - Secured a provisional patent for the invention

Awards

- 2026  **POPL 2026 Distinguished Paper Award**, *POPL Program Committee*
Awarded to up to 10% of accepted papers that demonstrate exceptional relevance, originality, significance, and clarity
- 2022  **Clarendon Scholarship**, *Clarendon Fund*
Funding awarded to outstanding graduate scholars
- 2019  **University Medal**, *University of Sydney*
Awarded to first place in the year
- 2018  **College Medal**, *St Andrews College*
Awarded to the highest academic achiever in the college
- 2014  **Gold Duke of Edinburgh Award**
Awarded to young adults for completing a series of multidisciplinary activities